THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE UNIT TEST SET 3

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The relation between Rhizobium and legume plant is an example of: 1

(a) symbiosis **(b)** predation **(c)** parasitism **(d)** saprotrophy

2. The ray of light falling on a surface is the: 1

(a) incident ray **(b)** reflected ray **(c)** emergent ray **(d)** normal

3. The phase when the Moon is not visible is: 1

(a) quarter **(b)** crescent **(c)** full moon **(d)** new moon

4. The oxygen we breathe is mostly from: 1

(a) burning **(b)** digestion
(c) photosynthesis **(d)** respiration of plants

5. The distance of the image from a plane mirror is ___ the distance of the object. 1

(a) unrelated to **(b)** half of **(c)** double **(d)** equal to

6. A lunar eclipse occurs when the ___ comes between the Sun and Moon. 1

(a) Earth **(b)** Venus **(c)** Mars **(d)** Jupiter

7. The raw materials for photosynthesis are: 1

(a) nitrogen and water **(b)** carbon dioxide and water
(c) starch and oxygen **(d)** oxygen and glucose

8. Which of these is non-luminous? 1

(a) a bulb **(b)** a candle flame **(c)** a star **(d)** the Moon

9. Day and night are caused by: 1

(a) revolution of the Earth **(b)** rotation of the Earth
(c) tilt of the Moon **(d)** revolution of the Moon

10. Loss of water vapour from the leaves is called: 1

(a) photosynthesis **(b)** digestion **(c)** respiration **(d)** transpiration

11. The partially dark region around the umbra is the: 1

(a) image **(b)** penumbra **(c)** umbra **(d)** focus

12. The Tropic of Cancer lies at: 1

(a) 66.5° N **(b)** 23.5° N **(c)** 0° **(d)** 23.5° S

13. Plants respire through: 1

(a) flowers only **(b)** stomata and roots

(c) leaves only **(d)** fruits only

14. The ray that bounces back is the: 1

(a) incident ray **(b)** reflected ray **(c)** refracted ray **(d)** normal

15. The phase when the whole disc is visible is: 1

(a) full moon **(b)** crescent **(c)** new moon **(d)** half

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. What are transparent, translucent and opaque materials? 2

17. How are day and night caused? 2

18. Name the modes of nutrition other than autotrophic in plants. 2

19. What is lateral inversion? 2

20. What is an equinox? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain why we have a leap year. 3

22. State the differences between photosynthesis and respiration. 3

23. Why does the Moon not produce its own light? 3

24. Explain the difference between solar and lunar eclipses. 3

OR

Explain how water and minerals move up from the roots to the leaves. 3

25. Describe the construction of a pinhole camera. 3

OR

Explain why the days are longer in summer than in winter. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe how the tilt of the Earth's axis causes seasons with the positions of June and December solstices. 5

OR

Explain the importance of plants for the ecosystem. 5

27. Discuss luminous and non-luminous objects with five examples and how we see non-luminous objects. 5

OR

Describe the effect of Earth's rotation on the direction of the Sun's apparent movement and time zones (basic). 5

END OF QUESTION PAPER — **Total: 50 Marks**